4.7 Lipid Catabolism

Microorganisms frequently use lipids such as triglyceride or triacylglycerol (esters of glycerol and fatty acids) as common reserve energy sources. These can be hydrolyzed to glycerol and fatty acid by microbial lipases. The glycerol is then phosphorylated and oxidized to Dihydroxyacetone phosphate and then catabolized in the Glycolysis pathway. Fatty acids from triacylglycerols and other lipids are often oxidized in the β-oxidation pathway. In this pathway fatty acids are degraded to acetyl CoA (2C segment), then it enters into the TCA cycle.



Bubble Test: Bubbles are a positive result for the presence of catalase. If an organism can produce catalase, it will produce bubbles of oxygen when hydrogen peroxide is added to it.

Infobits

One of the major environmental problems today is hydrocarbon contamination resulting from the activities related to the petrochemical industry. Several bacteria can use hydrocarbon as a feed and reduce pollution.

Pseudomonas putida (Super Bug) *Alcanivorax borkumens, Mycobacterium, Brevibacterium, Aspergillus, Penicillium, Candida lipolytica* are the most active agents in petroleum degradation and they work as primary degraders in oil spilled environment. These organisms are mainly in involved in bioremediation which reduce environmental pollution

4.8 Protein Catabolism

Many microbes use protein as their source of carbon and energy. Pathogenic microorganisms secrete protease enzyme that hydrolyze proteins and polypeptides to amino acids which are then transported into the cell and catabolized. Protease (Peptidase or proteinase) helps in proteolysis (Figure 4.8). These proteolytic enzymes break the long chains of proteins into peptides and eventually into amino acids. The enzymes are classified based on the sites at which they catalase the cleavage of proteins as exopeptidase and endopeptidase.

The protein catabolism involves two reactions namely,

- Deamination and
- Transamination

Deamination is the removal of the amino group from an amino acid. Transamination is the transferring of amino group from an amino acid to an amino acid acceptor.

The organic acid resulting from deamination can be converted to pyruvate, acetyl CoA or TCA cycle intermediates to release energy. Excess nitrogen from deamination may be excreted as ammonium ion.



Various metabolic processes such as blood coagulation, fibrinolysis, complement activation, phagocytosis and blood pressure control are regulated by proteases.

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